

An Exact Sparse Realization of the E_8 Cartan 3-Form

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September 2026

Abstract

The Cartan 3-form of a semisimple Lie algebra is the alternating trilinear form

$$\Omega(X, Y, Z) = B(X, [Y, Z]),$$

where B is an invariant nondegenerate bilinear form. We give an exact sparse realization of this form for the Chevalley $\mathbb{Q}$ -form of $\mathfrak{e}_8$, in a fixed ϵ -canonical Chevalley basis and a fixed normalization B_0 . Among the $248^3 = 15,252,992$ ordered basis triples, exactly 16,176 have nonzero coefficient: 13,440 in the root–root–root sector and 2,736 in the Cartan–root–root sector. The complete nonzero coefficient spectrum is

$$-2 : 24, \quad -1 : 8064, \quad +1 : 8064, \quad +2 : 24.$$

On root vectors, the support is exactly the zero-sum ternary incidence relation

$$\Omega(e_\alpha, e_\beta, e_\gamma) \neq 0 \iff \alpha + \beta + \gamma = 0,$$

which yields a 240-vertex, 2240-edge linear 3-uniform hypergraph representation of the root support. We derive the headline counts analytically, construct the full signed tensor by two algorithmically distinct exact routes, and validate it against an independently reconstructed E_8 bracket and invariant form. Jacobi is checked on all $\binom{248}{3} = 2,511,496$ distinct basis triples; the independently reconstructed map $B_0(X, [Y, Z])$ reproduces the sparse tensor entry-for-entry; and raising its first index with B_0^{-1} recovers all $248^2 = 61,504$ ordered basis brackets exactly. The underlying Lie-theoretic structures are classical; the contribution is their explicit sparse computational realization, complete coefficient census, root-incidence representation, deterministic machine-readable form, and reproducible exact validation architecture.

1 Introduction

The exceptional simple Lie algebra $\mathfrak{e}_8$ has rank 8, dimension 248, and a root system of 240 roots. Exact realizations of its multiplication are classical: Chevalley bases give integral structure constants, explicit E_8 commutation data have been tabulated, and modern formulas provide canonical structure constants in simply-laced types, including E_8 [1, 2]. Explicit formulas for the E_8 bracket also exist in alternative realizations [3], while recent public software provides sparse machine-readable structure constants [4].

Given an invariant nondegenerate symmetric bilinear form B , the Lie bracket can be lowered to the alternating trilinear form

$$\Omega(X, Y, Z) = B(X, [Y, Z]). \tag{1}$$

This Cartan 3-form is a classical object associated with a simple Lie algebra and has been studied geometrically and multisymplectically; see, for example, [5]. In mathematical physics the same lowering appears as a completely antisymmetric structure tensor, and explicit E_8 formulas in other bases occur in supergravity and related work [6]. Our purpose is therefore not to introduce

a new invariant. Instead, we ask for a complete exact sparse realization of the normalized E_8 Cartan 3-form in a fixed ϵ -canonical Chevalley basis, together with a transparent combinatorial description of its support, a deterministic machine-readable representation, and an independent exact validation chain.

Writing b_a for the basis elements fixed below, organizing the multiplication law as the scalar alternating tensor $g_{abc} = B_0(b_a, [b_b, b_c])$, rather than only as the vector-valued bracket C^a_{bc} , makes its support particularly transparent. On root vectors the support is precisely the zero-sum ternary relation

$$\Omega(e_\alpha, e_\beta, e_\gamma) \neq 0 \iff \alpha + \beta + \gamma = 0. \quad (2)$$

This gives 13,440 ordered root-sector coefficients, or 2240 unordered zero-sum triples. Equivalently, the root support is a 3-uniform hypergraph on 240 vertices with degree 28 and pair codegree 0 or 1. The counts 2240 and 1120 related A_2 classes are consistent with established E_8 combinatorics [7, 8, 9, 10]; no novelty is claimed for those counts themselves.

After the Cartan directions are included, the complete sparse support has

$$13,440 + 2,736 = 16,176 \quad (3)$$

ordered nonzero coefficients. In the basis and normalization fixed below, their multiplicities are

$$-2 : 24, \quad -1 : 8064, \quad +1 : 8064, \quad +2 : 24. \quad (4)$$

Both (3) and (4) admit short analytic derivations from the standard E_8 root inner-product census and alternation; the computer implementation independently reproduces those counts.

The full signed sparse tensor is constructed by two algorithmically distinct exact routes: one directly from ordered Lie brackets and invariant pairings, and one from canonical unordered root triples and opposite-root lines followed by alternating extension. The resulting sparse coefficient maps agree entry-for-entry. The final tensor is serialized deterministically, with exact integer coefficients and fixed basis/sign metadata.

For validation, the bracket and normalized bilinear form are then reconstructed independently of the tensor. The resulting algebra closes exactly, is antisymmetric, and satisfies Jacobi on all

$$\binom{248}{3} = 2,511,496$$

distinct basis triples. From this independently validated algebra we reconstruct $B_0(X, [Y, Z])$ and recover the sealed sparse tensor entry-for-entry. Conversely, because B_0 is nondegenerate, raising the first index of the trilinear tensor recovers the bracket; this equality is verified on all $248^2 = 61,504$ ordered basis pairs. Thus the sparse 3-form is not only a support representation but an exact lowered encoding of the complete E_8 multiplication law in the stated normalization.

The paper is organized as follows. Section 2 fixes the E_8 root, basis, sign, and bilinear-form conventions. Section 3 develops the zero-sum ternary incidence structure. Section 4 constructs and counts the sparse Cartan 3-form. Section 5 describes the two-route construction and independent algebraic validation. Section 6 specifies the machine-readable realization. Section 7 states the prior-art and contribution boundary explicitly, and Section 8 discusses computational consequences.

2 Conventions and the ϵ -canonical E_8 basis

We work first with the Chevalley $\mathbb{Q}$ -form $\mathfrak{g}_{\mathbb{Q}}$ of the split simple Lie algebra of type E_8 . All basis coefficients, structure constants, tensor entries, and validation identities are therefore exact over $\mathbb{Q}$. Scalar extension gives the corresponding split real algebra $\mathfrak{g}_{\mathbb{R}} = \mathfrak{g}_{\mathbb{Q}} \otimes_{\mathbb{Q}} \mathbb{R}$ or complex algebra $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g}_{\mathbb{Q}} \otimes_{\mathbb{Q}} \mathbb{C}$ without changing the coefficient formulas below. For notational economy we write $\mathfrak{e}_8$ when no distinction of ground field is relevant.

2.1 The root system

We use standard root-space conventions for semisimple Lie algebras [11]. We realize the E_8 root system $\Phi(E_8) \subset \mathbb{R}^8$ with the Euclidean inner product and root norm

$$(\alpha, \alpha) = 2.$$

The roots are the union of the 112 vectors obtained by permuting

$$(\pm 1, \pm 1, 0, 0, 0, 0, 0, 0)$$

and the 128 vectors

$$\frac{1}{2}(\pm 1, \dots, \pm 1)$$

with an even number of minus signs. Thus $|\Phi(E_8)| = 240$. For exact computation we internally use doubled coordinates $q = 2\alpha \in \mathbb{Z}^8$; then $(q, q) = 8$ for every root.

For a fixed root α , the standard E_8 inner-product census is

$$\#\{\beta : (\alpha, \beta) = -2, -1, 0, +1, +2\} = (1, 56, 126, 56, 1), \quad (5)$$

where the +2 entry is $\beta = \alpha$ and the -2 entry is $\beta = -\alpha$ [9]. This simple census drives several of the sparse-support formulas below.

2.2 Simple roots and Cartan matrix

We use the simple roots listed in doubled coordinates in Table 1. Dividing by two gives the corresponding roots in the norm-2 convention. Their Gram matrix is the E_8 Cartan matrix

$$A = \begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 0 & -1 \\ 0 & 0 & 0 & 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 2 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 2 \end{pmatrix}. \quad (6)$$

Its determinant is 1. Every root has an exact integral expansion

$$\alpha = \sum_{i=1}^8 n_i \alpha_i,$$

and we write

$$\text{ht}(\alpha) = \sum_i n_i.$$

For the chosen positive system there are 120 positive and 120 negative roots, and the heights range from -29 to 29.

2.3 ϵ -canonical Chevalley signs and invariant form

Let

$$\mathfrak{e}_8 = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Phi(E_8)} \mathfrak{g}_\alpha$$

Table 1: Frozen simple roots in doubled coordinates $q = 2\alpha$.

i	$2\alpha_i$
1	$(2, -2, 0, 0, 0, 0, 0, 0)$
2	$(0, 2, -2, 0, 0, 0, 0, 0)$
3	$(0, 0, 2, -2, 0, 0, 0, 0)$
4	$(0, 0, 0, 2, -2, 0, 0, 0)$
5	$(0, 0, 0, 0, 2, -2, 0, 0)$
6	$(0, 0, 0, 0, 0, 2, 2, 0)$
7	$(-1, -1, -1, -1, -1, -1, -1, -1)$
8	$(0, 0, 0, 0, 0, 2, -2, 0)$

be the root decomposition. We choose root vectors $e_\alpha \in \mathfrak{g}_\alpha$ using the ϵ -canonical Chevalley-basis formalism of Geck and Lang [2]. Relative to the simple-root ordering above, the frozen sign vector is

$$\epsilon = (+1, -1, +1, -1, +1, -1, +1, -1). \quad (7)$$

For $\alpha + \beta \in \Phi(E_8)$,

$$[e_\alpha, e_\beta] = N_{\alpha,\beta}^\epsilon e_{\alpha+\beta}, \quad N_{\alpha,\beta}^\epsilon \in \{-1, +1\}.$$

For opposite roots we use

$$[e_\alpha, e_{-\alpha}] = (-1)^{\text{ht}(\alpha)} h_\alpha, \quad h_\alpha = \sum_i n_i h_i. \quad (8)$$

We normalize the invariant symmetric bilinear form B_0 by

$$B_0(h_i, h_j) = a_{ij}. \quad (9)$$

Invariance together with (8) gives

$$B_0(e_\alpha, e_{-\alpha}) = (-1)^{\text{ht}(\alpha)}, \quad (10)$$

while distinct non-opposite root spaces are orthogonal and $\mathfrak{h}$ is orthogonal to all nonzero root spaces. The sign split in (10) is basis-dependent and is not an invariant of E_8 .

We order the basis as

$$\mathcal{B} = (h_1, \dots, h_8, e_{\alpha_0}, \dots, e_{\alpha_{239}}), \quad (11)$$

where the roots are sorted lexicographically in doubled coordinates. We write b_a for the a th element of $\mathcal{B}$. This fixes every coefficient reported below.

3 Zero-sum root incidence

3.1 Root support of the Cartan 3-form

Define

$$g(X, Y, Z) = B_0(X, [Y, Z]). \quad (12)$$

By invariance and antisymmetry of the Lie bracket, g is an alternating trilinear form, i.e.

$$g \in \Lambda^3(\mathfrak{e}_8)^*.$$

Proposition 3.1 (Root-sector support). *For $\alpha, \beta, \gamma \in \Phi(E_8)$,*

$$g(e_\alpha, e_\beta, e_\gamma) \neq 0 \iff \alpha + \beta + \gamma = 0.$$

Proof. If $\beta + \gamma$ is neither a root nor zero, then $[e_\beta, e_\gamma] = 0$. If $\gamma = -\beta$, then $[e_\beta, e_{-\beta}] \in \mathfrak{h}$, which pairs trivially with e_α . Otherwise, if $\beta + \gamma \in \Phi(E_8)$, then

$$[e_\beta, e_\gamma] \in \mathfrak{g}_{\beta+\gamma}.$$

The invariant form pairs $\mathfrak{g}_\alpha$ nontrivially only with $\mathfrak{g}_{-\alpha}$, so nonvanishing forces $\beta + \gamma = -\alpha$. Conversely, if $\alpha + \beta + \gamma = 0$, then $\beta + \gamma = -\alpha$ is a root, the bracket $[e_\beta, e_\gamma]$ is nonzero, and the opposite root spaces pair nondegenerately. $\square$

When the sum vanishes, the normalized coefficient is

$$g(e_\alpha, e_\beta, e_\gamma) = N_{\beta, \gamma}^\epsilon (-1)^{\text{ht}(\alpha)} \in \{-1, +1\}. \quad (13)$$

Thus the root-sector support is determined entirely by the zero-sum ternary relation, while the sign depends on the chosen canonical basis convention.

3.2 Support census

Proposition 3.2 (Ordered root-sector count). *The number of ordered nonzero root–root–root coefficients is*

$$|\text{supp}_{\text{RRR}} g| = 13,440.$$

Proof. Fix $\beta \in \Phi(E_8)$. By (5), there are exactly 56 roots γ with $(\beta, \gamma) = -1$. In a simply-laced root system this is equivalent to $\beta + \gamma \in \Phi(E_8)$. The third root is then uniquely $\alpha = -(\beta + \gamma)$. Therefore

$$|\text{supp}_{\text{RRR}} g| = 240 \cdot 56 = 13,440.$$

$\square$

Corollary 3.3. *There are*

$$\frac{13,440}{6} = 2240$$

unordered zero-sum root triples. Negation pairs these into 1120 classes.

Indeed, if $\alpha + \beta + \gamma = 0$, then the three roots have pairwise inner product -1 , so the six roots $\{\pm\alpha, \pm\beta, \pm\gamma\}$ form an A_2 subsystem. The negated triple gives the same subsystem, and conversely every A_2 subsystem contains exactly two zero-sum triples, exchanged by negation. Thus the 1120 negation classes are in bijection with the 1120 A_2 root subsystems of E_8 . The latter count is listed explicitly by Bordner, Manton, and Sasaki [8] and also derived directly from the $240 \cdot 56$ root-pair census in [10]. We use the zero-sum representation because it exposes the ternary support directly.

3.3 A 3-uniform root hypergraph

Let $\mathcal{H} = (V, E)$ have vertex set $V = \Phi(E_8)$ and edge set

$$E = \{\{\alpha, \beta, \gamma\} : \alpha + \beta + \gamma = 0\}.$$

Then $|V| = 240$ and $|E| = 2240$.

Proposition 3.4 (Regularity). *Every vertex of $\mathcal{H}$ has degree 28.*

Proof. Fix α . There are 56 roots β with $(\alpha, \beta) = -1$. Each hyperedge containing α contains exactly two such other roots, and either one determines the third root uniquely. Hence $2 \deg(\alpha) = 56$. $\square$

Proposition 3.5 (Linearity). *Any two distinct vertices lie in at most one hyperedge. Equivalently, pair codegrees are 0 or 1.*

Proof. If $\{\alpha, \beta, \gamma\}$ is an edge, then $\gamma = -(\alpha + \beta)$, so a pair determines at most one third root. $\square$

Consequently, the number of active unordered root pairs is

$$3|E| = 6720.$$

The hypergraph is therefore a convenient finite data representation of the unsigned root support of g , rather than a new root-system invariant.

4 The complete sparse Cartan 3-form

4.1 Sector structure

For the basis (11), nonzero coefficients occur only in two sector types:

- (i) three root vectors, with the zero-sum criterion of Proposition 3.1;
- (ii) one Cartan vector and an opposite pair $e_\alpha, e_{-\alpha}$.

All sectors containing two or three Cartan entries vanish.

For the second type,

$$g(h_i, e_\alpha, e_{-\alpha}) = (-1)^{\text{ht}(\alpha)} \alpha(h_i), \quad (14)$$

with the other five orderings obtained by alternation.

Proposition 4.1 (Cartan evaluation census). *For each simple Cartan direction h_i , exactly 114 roots satisfy $\alpha(h_i) \neq 0$.*

Proof. Because E_8 is simply laced, $\alpha(h_i) = (\alpha, \alpha_i)$. Applying (5) to the fixed root α_i , the values $-2, -1, 0, +1, +2$ occur with multiplicities 1, 56, 126, 56, 1. Hence the nonzero multiplicity is $1 + 56 + 56 + 1 = 114$. $\square$

Proposition 4.2 (Ordered Cartan–root–root count). *The number of ordered nonzero coefficients with one Cartan and two root arguments is*

$$|\text{supp}_{\text{HRR}} g| = 2736.$$

Proof. For one fixed Cartan position, Proposition 4.1 gives

$$8 \cdot 114 = 912$$

nonzero ordered coefficients. The Cartan argument can occupy any of the three positions, hence $3 \cdot 912 = 2736$. $\square$

Theorem 4.3 (Complete sparse support census). *In the frozen basis,*

$$|\text{supp } g| = 16,176.$$

Proof. The only nonzero sectors are RRR and HRR, so Propositions 3.2 and 4.2 give

$$|\text{supp } g| = 13,440 + 2,736 = 16,176.$$

$\square$

The complete ordered domain has size $248^3 = 15,252,992$, so the support density is

$$\frac{16,176}{15,252,992} \approx 1.0605 \times 10^{-3},$$

i.e. approximately 0.1061%.

4.2 Coefficient multiplicities

Proposition 4.4 (Root-sector sign balance). *Among the 13,440 nonzero root-sector coefficients,*

$$-1 : 6720, \quad +1 : 6720.$$

Proof. Every unordered zero-sum root triple generates six ordered permutations. Since g is alternating, three even permutations have coefficient c and three odd permutations have coefficient $-c$. By (13), $|c| = 1$. Thus each of the 2240 unordered triples contributes three coefficients of each sign. $\square$

Proposition 4.5 (Magnitude-two Cartan sector). *Exactly 48 ordered HRR coefficients have magnitude 2, split as*

$$-2 : 24, \quad +2 : 24.$$

Proof. For fixed i , the only roots with $|\alpha(h_i)| = 2$ are $\pm\alpha_i$. Hence each simple Cartan direction gives one underlying unordered basis triple

$$\{h_i, e_{\alpha_i}, e_{-\alpha_i}\}.$$

There are eight such triples and six ordered permutations of each, giving 48 magnitude-two coefficients. Alternation splits them equally by sign. $\square$

The remaining $2736 - 48 = 2688$ HRR coefficients have magnitude 1, hence 1344 of each sign.

Theorem 4.6 (Complete coefficient spectrum). *The complete nonzero coefficient spectrum is*

$-2 : 24,$	$-1 : 8064,$	$+1 : 8064,$	$+2 : 24.$
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Proof. Combine Proposition 4.4 with the HRR contributions

$$-2 : 24, \quad -1 : 1344, \quad +1 : 1344, \quad +2 : 24.$$

Then $6720 + 1344 = 8064$ for each unit sign. $\square$

Remark 4.7. The spectrum in Theorem 4.6 is a census in the fixed basis and normalization, not an intrinsic spectrum of E_8 . Rescaling root vectors or the invariant form changes coefficient values while leaving the abstract Cartan 3-form unchanged up to the corresponding normalization.

5 Independent construction and validation

5.1 Two algorithmically distinct exact tensor constructions

The complete sparse tensor was constructed by two algorithmically distinct exact routes. They share the frozen basis, sign, and normalization conventions, but differ in how the signed support is generated.

Route A: direct ordered construction. For every ordered root pair (β, γ) with $\beta + \gamma \in \Phi(E_8)$, set $\alpha = -(\beta + \gamma)$ and evaluate

$$g(e_\alpha, e_\beta, e_\gamma) = N_{\beta, \gamma}^\epsilon B_0(e_\alpha, e_{-\alpha}).$$

For the Cartan sector, evaluate (14) separately for each Cartan position and each oriented opposite-root pair. No permutation expansion is used to generate the primary map.

Route B: alternating reconstruction. Begin with the 2240 canonical unordered zero-sum root triples and the canonical opposite-root lines. Evaluate one ordering of each underlying nonzero basis triple and generate the other five orderings only by permutation parity.

The resulting sparse maps contain 16,176 entries each and agree exactly:

comparison	discrepancies
direct-only keys	0
reconstruction-only keys	0
shared keys with unequal coefficients	0

This is stronger than agreement of aggregate counts: support and signed coefficients coincide entry-for-entry.

5.2 Independent reconstruction of the Lie algebra

To validate the tensor without using the tensor itself to define the algebra, the full bracket was reconstructed independently from the root arithmetic, canonical structure constants, Cartan-root evaluations, and opposite-root brackets. Over the $248^2 = 61,504$ ordered basis-pair domain, exactly 15,504 pairs have nonzero bracket. This counts nonzero ordered input pairs (i, j) , not individual nonzero scalar structure-constant entries C^k_{ij} : an opposite-root bracket can have several Cartan components. Thus this basis-pair census is distinct from sparse scalar-entry counts such as the 16,694 nonzero f_{ij}^k entries reported by public implementations. The bracket closes in the frozen basis, is antisymmetric, and has zero self-brackets.

The Jacobiator

$$J(X, Y, Z) = [X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]]$$

was then evaluated on every canonical triple of distinct basis elements. Since antisymmetry of the bracket has already been established, the Jacobiator is alternating; triples with repeated basis elements therefore vanish identically. It is consequently sufficient for exhaustive basis-level verification to check the $\binom{248}{3}$ distinct triples. Table 2 gives the complete sector census.

Table 2: Exhaustive Jacobi audit.

sector	triples checked	failures
HHH	56	0
HHR	6,720	0
HRR	229,440	0
RRR	2,275,280	0
total	2,511,496	0

The bilinear form B_0 was also reconstructed independently. Its sparse basis matrix contains 262 nonzero ordered entries with nonzero value census

$$-1 : 142, \quad +1 : 112, \quad +2 : 8.$$

It is symmetric, its Cartan block has determinant 1, and the opposite-root pairings make the root block nondegenerate.

Two sparse trilinear maps were then formed independently,

$$L(X, Y, Z) = B_0([X, Y], Z), \quad R(X, Y, Z) = B_0(X, [Y, Z]).$$

Each contains 16,176 nonzero entries, and their exact comparison has zero key or coefficient discrepancies. Hence the invariant-form identity is recovered computationally without reference to the previously constructed tensor.

Finally, the algebra-derived map R was compared with the already sealed sparse tensor. Both support and coefficients agree entry-for-entry.

5.3 Recovering the bracket from g

The sparse tensor is not merely a support detector. Because B_0 is nondegenerate, it contains the full multiplication law after raising one index.

Proposition 5.1 (Bracket recovery). *For all $Y, Z \in \mathfrak{e}_8$,*

$$[Y, Z] = B_0^{-1}(g(\cdot, Y, Z)).$$

In components,

$$C^a{}_{bc} = B_0^{ad} g_{dbc},$$

where $B_{0,ad} := B_0(b_a, b_d)$ and B_0^{ad} denotes the inverse matrix.

Proof. By definition, $g(X, Y, Z) = B_0(X, [Y, Z])$ for every X . Nondegeneracy of B_0 gives a unique vector W such that $B_0(X, W) = g(X, Y, Z)$ for all X . The vector $W = [Y, Z]$ satisfies this condition. $\square$

In the implementation, Proposition 5.1 was checked on all 61,504 ordered basis pairs. The recovered bracket and the independently frozen bracket had zero support or coefficient discrepancies.

6 Machine-readable realization and reproducibility

All computational stages use exact integer or rational arithmetic. Roots are stored in doubled integral coordinates, the sparse tensor coefficients are integers, and arbitrary test vectors use exact rational coefficients. No floating-point threshold is used to decide support or equality.

The canonical sparse artifact contains one record for every nonzero ordered basis coefficient and includes:

- the 248-element basis ordering;
- the 240 doubled root coordinates;
- the canonical sign vector (7);
- the normalization $B_0(h_i, h_j) = a_{ij}$;
- the sector label RRR or HRR;
- the exact integer coefficient.

The sealed artifact contains 16,176 entries and has SHA-256

43b52b4116dfcf0dace332234b5c9dc50d95d51effb690aec8aebc39ad4f92e8.

The source code, deterministic data artifact, validation suite, and reproducibility records are publicly available in the `e8-ternary-form-audit` repository at <https://github.com/salaealer9/e8-ternary-form-audit>. The frozen reproducibility release is v1.0.0, commit 2000d93478be897f0742191aafd173d71e481fbd, archived on Zenodo at <https://doi.org/10.5281/zenodo.22676423>. The canonical sparse tensor artifact `data/derived/phase3d_signed_`

`sparse_cartan_form.json` has SHA-256 43b52b4116dfcf0dace332234b5c9dc50d95d51effb690aec8aeb39ad4f92e8.

The development workflow additionally uses signed version-control checkpoints and deterministic hash manifests so that protocol freezes, primary constructions, validation stages, and final artifacts are chronologically distinguishable. The mathematical results in Sections 3–4, however, are stated independently of that workflow and can be checked analytically.

7 Prior art and contribution boundary

The Cartan 3-form itself is classical [5]. Fully antisymmetric lowered E_8 structure constants have long been used in mathematical physics [6]. Explicit Chevalley constants for E_8 appear in Đoković’s appendix [1], and modern canonical formulas for simply-laced simple Lie algebras are given by Geck and Lang [2]. Explicit formulas for the E_8 bracket are also available in alternative realizations [3]. Recent public software provides machine-readable sparse E_8 brackets with 16,694 nonzero scalar structure-constant entries C^k_{ij} [4]. This scalar-entry count is not the same object as the 15,504 nonzero ordered basis-pair count used in Section 5.

The root combinatorics used here are likewise classical. The E_8 root inner-product graph has been studied systematically [9]; ternary/ A_2 models of exceptional Lie algebras appear in Manivel’s work [7]; and the exact count of 1120 A_2 root subsystems appears explicitly in both older and recent literature [8, 10].

Accordingly, we do not claim a new E_8 invariant, a new Lie bracket, a new Chevalley basis, or discovery of the 2240 zero-sum triples or 1120 A_2 classes. A dedicated prior-art audit found no equivalent complete realization in the sources searched for the combination presented here: the deterministic 16,176-entry lowered Cartan 3-form in the stated basis and normalization, its complete coefficient census, the zero-sum root hypergraph as an explicit support data object, the two algorithmically distinct exact reconstruction routes, and the independent exhaustive sealed-artifact validation pipeline. This is therefore best viewed as a computational/data/reproducibility contribution built from classical Lie-theoretic ingredients, rather than as a new theorem-level invariant.

We deliberately avoid “first” or “previously unknown” claims: failure to locate an equivalent artifact is not proof of universal absence from the literature.

8 Discussion

The sparse realization makes three aspects of the Cartan 3-form especially transparent.

First, the root sector is combinatorial. The unsigned support is exactly the zero-sum hypergraph on $\Phi(E_8)$, while the canonical Chevalley signs orient those admissible triples by coefficients ± 1 . In that sense, the root-sector tensor simultaneously provides an incidence predicate and a signed orientation.

Second, adding the Cartan sector turns this incidence skeleton into the complete basis-level Cartan 3-form. The extra 2736 coefficients are precisely the simple-Cartan evaluations on opposite-root lines. The appearance of magnitudes 2 is fully controlled by the simple roots, while all other nonzero coefficients have magnitude 1 in the chosen normalization.

Third, the sparse tensor is an alternative computational representation of the multiplication law. Together with B_0 , it determines the bracket by Proposition 5.1. Depending on a downstream task, one can therefore choose between the sparse bracket C^a_{bc} and the fully alternating lowered tensor g_{abc} . The latter can be useful when permutation symmetry and scalar contraction are more natural than vector-valued multiplication.

The present paper intentionally stops at this representation layer. Possible uses in invariant numerical schemes, exceptional-symmetry models, graph/hypergraph algorithms, or mathematical

physics require separate problem-specific analysis. No physical application is inferred merely from the existence of the sparse tensor.

9 Conclusion

We have given an exact sparse realization of the normalized E_8 Cartan 3-form

$$g(X, Y, Z) = B_0(X, [Y, Z])$$

in a fixed ϵ -canonical Chevalley basis. The root-sector support is exactly the zero-sum ternary relation on the 240 roots, giving 13,440 ordered root-sector entries or 2240 unordered hyperedges. The Cartan-root-root sector contributes 2736 further ordered entries, for a total sparse support of

$$16,176.$$

The complete nonzero coefficient multiplicities are

$$-2 : 24, \quad -1 : 8064, \quad +1 : 8064, \quad +2 : 24.$$

These headline counts were derived analytically and independently reproduced computationally.

Two exact tensor constructions agree entry-for-entry. An independent reconstruction of the Lie bracket and invariant form satisfies Jacobi on all 2,511,496 distinct basis triples and reproduces the sealed sparse tensor exactly. Conversely, raising the first tensor index with B_0^{-1} recovers all 61,504 ordered basis brackets. Thus the machine-readable 3-form is a complete exact lowered representation of the E_8 multiplication law in the stated normalization.

The underlying Lie-theoretic ingredients are classical. The contribution of the present work is their explicit sparse realization, complete coefficient census, root-incidence packaging, deterministic data representation, and reproducible cross-validation architecture.

A Basis and combinatorial summary

For convenience, Table 3 collects the principal finite counts used in the paper.

Table 3: Principal exact counts.

quantity	count
E_8 roots	240
unordered zero-sum root triples	2240
ordered RRR support	13,440
A_2 /negation classes	1120
hypergraph degree	28
active unordered root pairs	6720
ordered HRR support	2736
total nonzero g_{ijk}	16,176
full ordered tensor domain	15,252,992
nonzero ordered bracket pairs	15,504
distinct basis triples checked for Jacobi	2,511,496
ordered basis pairs checked for bracket recovery	61,504

B Coefficient formulas

For ease of implementation, the two nonzero sector formulas are

$$g(e_\alpha, e_\beta, e_\gamma) = N_{\beta, \gamma}^\epsilon (-1)^{\text{ht}(\alpha)} \quad \text{if } \alpha + \beta + \gamma = 0,$$

and

$$g(h_i, e_\alpha, e_{-\alpha}) = (-1)^{\text{ht}(\alpha)} \alpha(h_i),$$

with all other orderings determined by alternation. All HHH and HHR sectors vanish.

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